

Experiment 3

Name:Pratik Chavan

Div/Batch:A/A1 Roll No.07

.MODEL SMALL

.STACK 100H

.DATA

 ARRAY DB 10, 25, 15, 40, 5, 30, 50, 20

 LEN EQU \$ - ARRAY

 MIN DB 0

 MAX DB 0

.CODE

MAIN PROC

 MOV AX, @DATA

 MOV DS, AX

 MOV SI, 0

 MOV AL, ARRAY[SI]

 MOV MIN, AL

 MOV MAX, AL

FIND_MIN_MAX:

 MOV AL, ARRAY[SI]

 CMP AL, MAX

 JG UPDATE_MAX

 CMP AL, MIN

JL UPDATE_MIN

JMP NEXT_ELEMENT

UPDATE_MAX:

MOV MAX, AL

JMP NEXT_ELEMENT

UPDATE_MIN:

MOV MIN, AL

NEXT_ELEMENT:

INC SI

CMP SI, LEN

JL FIND_MIN_MAX

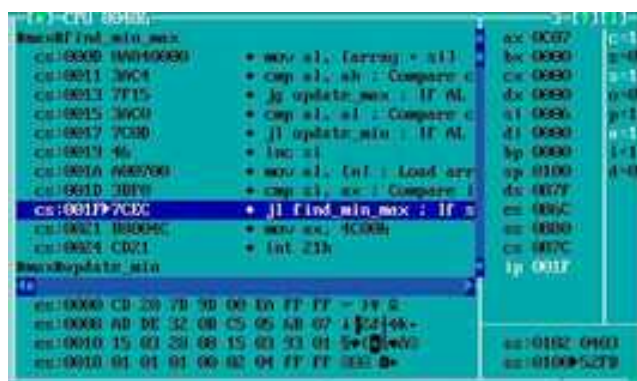
MOV AX, 4C00H

INT 21H

MAIN ENDP

END MAIN

OUTPUT:



The screenshot shows a debugger window with the following content:

- Disassembly:**
 - cs:0000:0000 - mov al, [array + al]
 - cs:0001:0004 - cmp al, ah ; Compare c
 - cs:0002:0008 - jg update_max ; If >
 - cs:0003:000C - cmp al, al ; Compare c
 - cs:0004:0010 - jl update_min ; If <
 - cs:0005:0014 - inc al
 - cs:0006:0018 - mov al, [nt] ; Load arr
 - cs:0007:001C - cmp al, ax ; Compare l
 - cs:0008:0020 - jl find_min_max ; If <
 - cs:0009:0024 - mov ax, 4C00H
 - cs:000A:0028 - int 21h
- Registers:**
 - ax: 0007
 - bx: 0000
 - cx: 0000
 - dx: 0000
 - si: 0000
 - di: 0000
 - bp: 0000
 - sp: 0100
 - ip: 007F
- Memory:**
 - 0000:0000 - CD 20 7B 5D 00 00 FF FF - 1F 0
 - 0000:0004 - 0B 0E 32 0B C5 05 5B 07 - 1F 0
 - 0000:0008 - 15 03 2B 0B 15 03 53 01 - 5F 0
 - 0000:000C - 01 01 01 00 02 04 FF FF - 00 0